

**Linear Algebra and Differential Equations**  
**Home Assignment-5**

Instructions for solving the assignment:

- i) Every student is supposed to solve 3 sub - question from each of the main questions (i.e., total 3 questions) based on the following rule:  
 Consider last 3 digits of your GR number and solve the corresponding sub questions.  
 Example: If your GR number is “12411309” then the student will solve sub-question 3 of Question A, sub-question 0 of Question B and sub-question 9 of Question C
- ii) Write the assignment in the section for Home assignments in your tutorial notebook.
- iii) Get the assignment checked by your tutorial teacher on or before 22.05.2025

**Q.A] Find the Eigen values and Eigen vectors of the following matrix**

**State Algebraic Multiplicity, Geometric Multiplicity and Eigen Space for each Eigen value.**

0)  $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$

1)  $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

2)  $A = \begin{bmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{bmatrix}$

3)  $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$

4)  $A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$

5)  $A = \begin{bmatrix} 7 & 0 & -2 \\ 0 & 5 & -2 \\ -2 & -2 & 6 \end{bmatrix}$

6)  $A = \begin{bmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{bmatrix}$

7)  $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$

8)  $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$

9)  $A = \begin{bmatrix} 7 & -2 & -2 \\ -2 & 1 & 4 \\ -2 & 4 & 1 \end{bmatrix}$

**Q.B] Diagonalize the following matrix:**

0)  $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$

1)  $A = \begin{bmatrix} 8 & 4 \\ 2 & 1 \end{bmatrix}$

2)  $A = \begin{bmatrix} 7 & 1 \\ 2 & 6 \end{bmatrix}$

3)  $A = \begin{bmatrix} 6 & 3 \\ 1 & 4 \end{bmatrix}$

4)  $A = \begin{bmatrix} 6 & 2 \\ 1 & 5 \end{bmatrix}$

5)  $A = \begin{bmatrix} 8 & 2 \\ 1 & 7 \end{bmatrix}$

6)  $A = \begin{bmatrix} 1 & 4 \\ 6 & 3 \end{bmatrix}$

7)  $A = \begin{bmatrix} 10 & 1 \\ 3 & 8 \end{bmatrix}$

8)  $A = \begin{bmatrix} 9 & 4 \\ -1 & 4 \end{bmatrix}$

9)  $A = \begin{bmatrix} 6 & 2 \\ 1 & 5 \end{bmatrix}$

**Q.C] Convert the quadratic form,  $Q(X) = Q(x_1, x_2)$  into canonical form  $Q(Y) = Q(y_1, y_2)$  using orthogonal diagonalization and hence identify the conic section,  $Q(y_1, y_2) = 1$**

0)  $Q(x_1, x_2) = 5x_1^2 + 4x_1x_2 + 5x_2^2$

1)  $Q(x_1, x_2) = 4x_1^2 + 12x_1x_2 + 9x_2^2$

2)  $Q(x_1, x_2) = x_1^2 + 4x_1x_2 + 4x_2^2$

3)  $Q(x_1, x_2) = x_1^2 - 6x_1x_2 + 9x_2^2$

4)  $Q(x_1, x_2) = 5x_1^2 + 6x_1x_2 + 5x_2^2$

5)  $Q(x_1, x_2) = 2x_1^2 + 8x_1x_2 + 8x_2^2$

6)  $Q(x_1, x_2) = 6x_1^2 + 4x_1x_2 + 3x_2^2$

7)  $Q(x_1, x_2) = 4x_1^2 - 4x_1x_2 + x_2^2$

8)  $Q(x_1, x_2) = 10x_1^2 - 4x_1x_2 + 7x_2^2$

9)  $Q(x_1, x_2) = 7x_1^2 + 4x_1x_2 + 7x_2^2$